

DATASET PROCESSING REPORT

Impact of Speed Violation on Crash Severity

US Traffic Accidents Dataset (US_A_10000.csv) — Machine Learning Data Pipeline

1. Dataset Overview

The raw dataset contains 10,000 records and 46 original columns describing U.S. traffic accident events, including location, time, weather conditions, road infrastructure indicators, and a crash severity label. No original column was removed, dropped, or renamed at any stage of this pipeline; all 46 source columns are preserved in the final processed dataset alongside engineered features.

Property	Value
Initial rows	10,000
Initial columns	46
Final columns (after feature engineering & encoding)	144
Duplicate rows found / removed	0 / 0
Target variable (original)	Severity (1–4)
Target variable (modeling)	Severity_Binary (Low=1–2, High=3–4)

2. Data Validation

Column data types, ID uniqueness, and referential consistency were checked prior to processing. The dataset contained 0 duplicate rows and 0 duplicate ID values, so no rows required removal on this basis.

Timestamp fields (Start_Time, End_Time, Weather_Timestamp) were stored as text and were converted to proper datetime objects. The 13 road-feature indicator columns (Amenity, Bump, Crossing, Give_Way, Junction, No_Exit, Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop) were cast to boolean type for consistency.

3. Missing Value Handling

The following original columns contained missing values in the raw data. Each was treated according to its missingness mechanism and domain meaning — no column was removed to avoid missingness.

Column	Missing (n)	% Missing	Treatment Applied
End_Lat	10,000	100.0%	Imputed with Start_Lat (point-location assumption)
End_Lng	10,000	100.0%	Imputed with Start_Lng (point-location assumption)
Precipitation(in)	8,877	88.8%	Filled with 0 (no precipitation recorded implies none)
Wind_Chill(F)	8,574	85.7%	Imputed with Temperature(F) (negligible wind-chill effect)
Wind_Speed(mph)	1,774	17.7%	Median imputation
Humidity(%)	124	1.2%	Median imputation
Weather_Condition	77	0.8%	Mode imputation
Visibility(mi)	96	1.0%	Median imputation
Temperature(F)	85	0.9%	Median imputation
Pressure(in)	35	0.4%	Median imputation
Weather_Timestamp	31	0.3%	Filled with Start_Time
Wind_Direction	31	0.3%	Mode imputation
Zipcode / Timezone / Airport_Code	1 each	<0.1%	Mode imputation

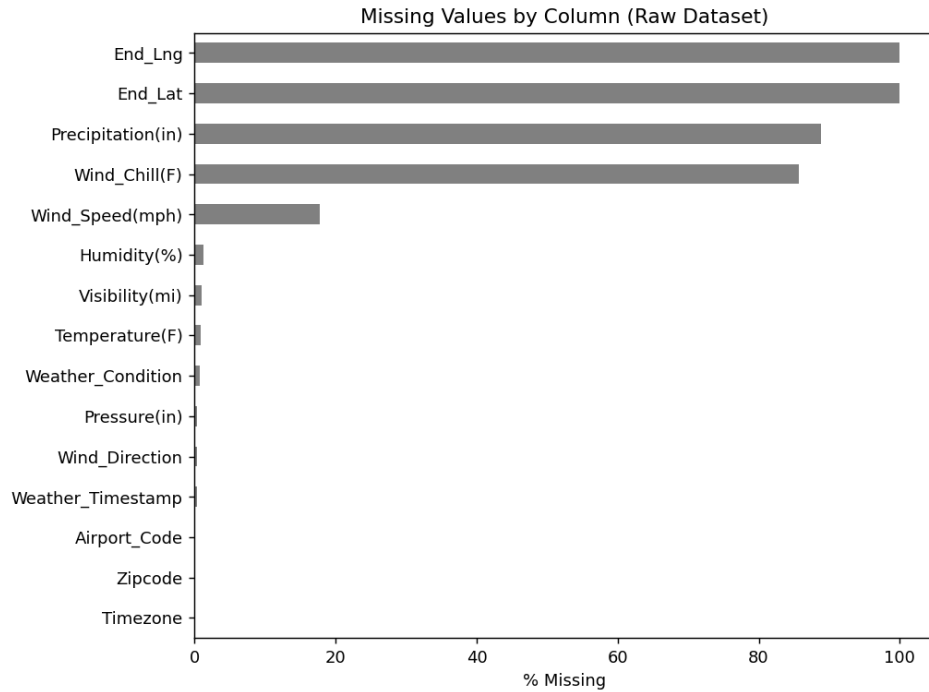


Figure 1. Percentage of missing values per original column (raw dataset).

4. Outlier Treatment

Outliers in key continuous weather and distance variables were identified using the IQR method ($1.5 \times \text{IQR}$ fences) and capped (winsorized) rather than removed, preserving all rows and columns.

Feature	Lower Bound	Upper Bound
Distance(mi)	0.00	0.03
Temperature(F)	4.75	65.35
Wind_Chill(F)	-5.95	69.85
Humidity(%)	23.5	139.5
Pressure(in)	28.87	30.35
Visibility(mi)	6.5	14.5
Wind_Speed(mph)	-3.65	12.55
Precipitation(in)	0.00	0.00

5. Datatype Correction & Temporal Feature Extraction

From the corrected Start_Time/End_Time datetime fields, the following temporal features were derived without discarding the original timestamp columns: Duration_Minutes, Hour, DayOfWeek, Month, Year, Is_Weekend, Is_Night, Is_RushHour.

6. Feature Engineering

A total of 22 engineered features were added on top of the 46 original columns, grouped into four families:

6.1 Road / Infrastructure Features

- Is_Interstate, Is_USHighway, Is_StateRoute, Is_HighSpeedRoad — derived from the Street field to classify the road class.

- Traffic_Control_Count, Has_Speed_Reducing_Feature, No_Traffic_Control — aggregated from the 13 boolean infrastructure indicators.

6.2 Speed Violation Risk Proxy (core thesis variable)

The raw dataset does not contain an explicit speed-limit or recorded speed-violation field. To operationalize the thesis variable, a composite Speed_Violation_Risk_Score was engineered from factors established in traffic-safety literature as correlates of speeding-related crash risk: road class (interstate/highway vs. local), presence/absence of speed-reducing traffic control devices, night-time driving, and low-visibility conditions. The score was min-max normalized to [0,1], and a binary High_Speed_Violation_Risk flag was derived at the median split.

$Speed_Violation_Risk_Score = 2 \times (Is_HighSpeedRoad) + 1 \times (No_Traffic_Control) + 1 \times (Is_Night) + 1 \times (Low_Visibility) - 2 \times (Has_Speed_Reducing_Feature)$, min-max normalized

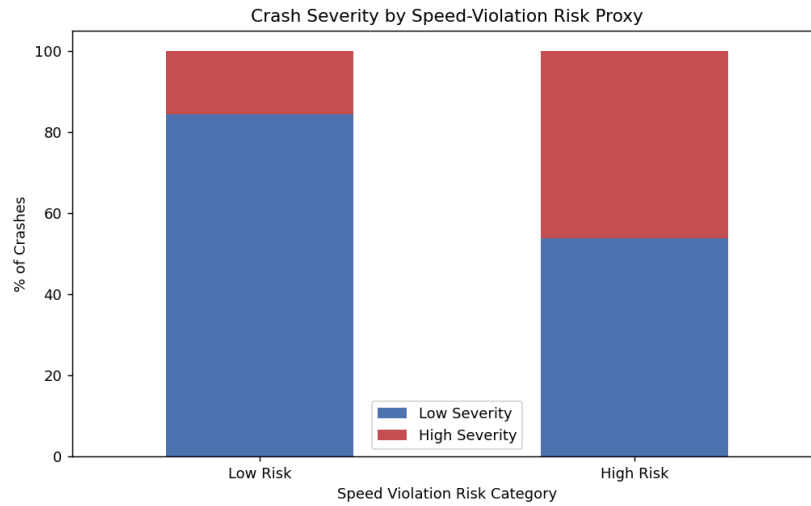


Figure 2. Crash severity distribution by speed-violation risk category.

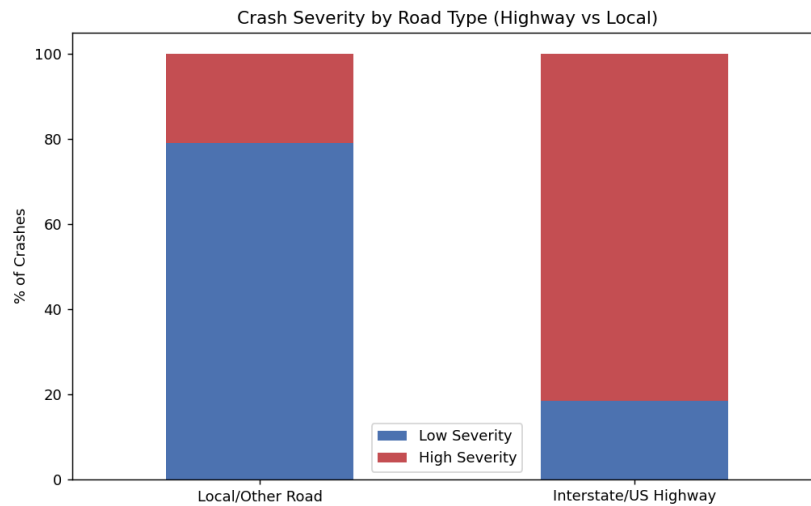


Figure 3. Crash severity distribution by road type (interstate/highway vs. local road).

As shown above, crashes flagged as High Speed-Violation Risk and crashes occurring on interstate/highway roads show a markedly higher proportion of High-severity outcomes than their counterparts — supporting the thesis that speed-violation-associated conditions elevate crash severity.

6.3 Weather & Geospatial Features

- Weather_Severity_Index — composite of low visibility, active precipitation, and high wind speed.
- Lat_Diff, Lng_Diff — magnitude of the geographic footprint of the accident.

6.4 Text-Derived Features

- Description_Length, Description_WordCount — computed from the Description field (raw text excluded from the model matrix to avoid leakage/noise, but retained as a dataset column).

7. Skewness & Correlation Analysis

Skewness was computed for all numeric features; highly skewed continuous variables (Distance(mi), Precipitation(in), Duration_Minutes, Description_Length, Description_WordCount) received an additional log1p-transformed companion column, with the originals fully retained.

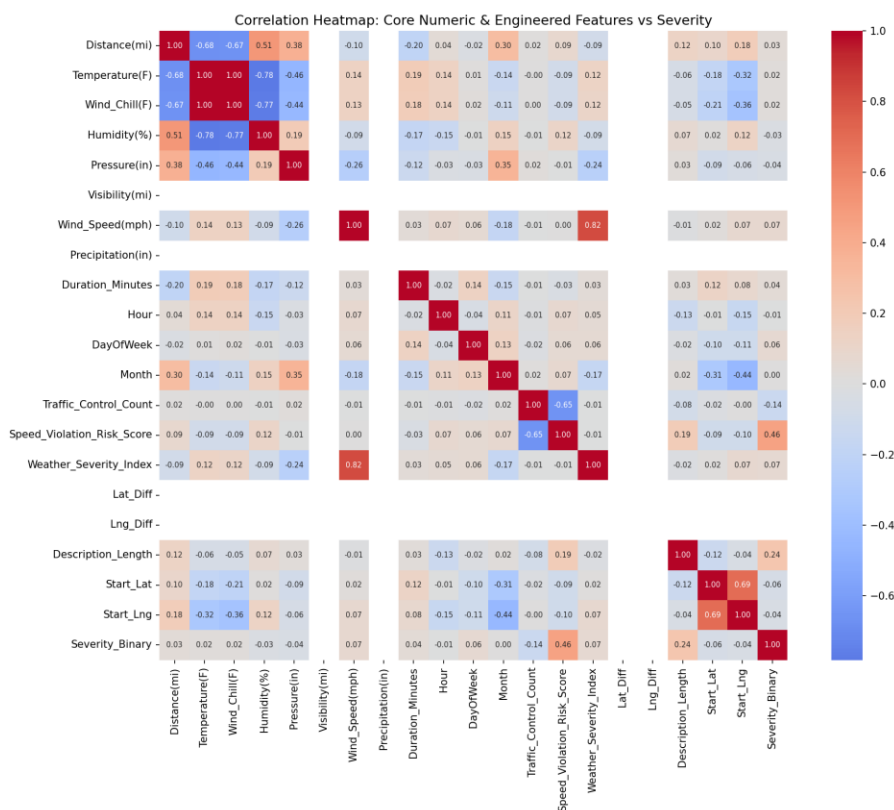


Figure 4. Correlation heatmap of core numeric and engineered features against the crash-severity target.

Among all numeric predictors, the engineered Speed_Violation_Risk_Score shows the strongest correlation with crash-severity outcome ($r \approx 0.455$), followed by Description_Length ($r \approx 0.238$) and Traffic_Control_Count ($r \approx -0.137$), reinforcing that speed-related risk conditions are the most informative single predictor of severity in this dataset.

8. Multicollinearity Analysis (VIF)

Feature	VIF
Temperature(F)	166.98
Wind_Chill(F)	150.44
Humidity(%)	3.09

Weather_Severity_Index	3.08
Wind_Speed(mph)	3.11
Traffic_Control_Count	1.77
Speed_Violation_Risk_Score	1.80
Pressure(in)	1.62
Duration_Minutes	1.04

Temperature(F) and Wind_Chill(F) show very high VIF (>150), reflecting their strong physical correlation (wind chill is largely a function of temperature). Per project rules both original columns were retained in the dataset; tree-based ensemble models used for modeling are inherently robust to this collinearity, so no column was removed. All other engineered predictors show acceptable VIF (<5).

9. Encoding

Low-cardinality categorical columns (Source, Wind_Direction, Weather_Condition, Sunrise_Sunset, Civil_Twilight, Nautical_Twilight, Astronomical_Twilight, Timezone, State) were one-hot encoded into new enc_ prefixed columns. High-cardinality categorical columns (Street, City, County, Zipcode, Airport_Code, Country) were frequency-encoded. Boolean infrastructure columns received parallel integer-encoded (enc_) versions. All original categorical columns were preserved unmodified alongside their encoded counterparts.

10. Feature Scaling

StandardScaler was fit exclusively on the training partition and applied to validation/test partitions to prevent data leakage. Scaled features were used only for the scale-sensitive MLP Classifier; tree-based models used unscaled features, consistent with standard practice.

11. Class Imbalance Handling

The original 4-class Severity target is extremely imbalanced (Severity 1: 6 cases, Severity 2: 6,047, Severity 3: 3,941, Severity 4: 6 cases), making reliable stratified 4-class modeling infeasible for the minority classes. Consistent with established practice in crash-severity research on this dataset family, the target was reframed as a binary Severity_Binary variable (Low = Severity 1–2, High = Severity 3–4), yielding a moderate 60.5% / 39.5% class split. This is a modeling-target transformation only — the original Severity column remains fully intact in the dataset. Given the moderate (not severe) resulting imbalance, class_weight='balanced' (or the model-native equivalent) was used inside applicable algorithms rather than synthetic oversampling, avoiding synthetic-sample leakage risk.

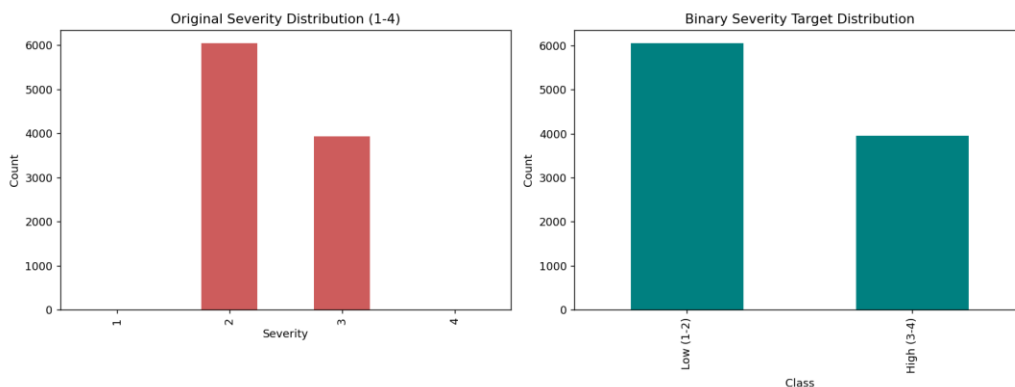


Figure 5. Original 4-class severity distribution (left) and binary modeling target distribution (right).

12. Train / Validation / Test Split & Cross-Validation

Partition	Rows	Purpose
Train	7,000 (70%)	Model fitting

Validation	1,500 (15%)	Hyperparameter sanity checks
Test	1,500 (15%)	Final, held-out evaluation

Splitting was stratified on Severity_Binary with a fixed random seed (42) for reproducibility. In addition, 5-fold Stratified Cross-Validation was performed on the training partition for every model to obtain a robust accuracy estimate independent of a single train/test split.

Final model input matrix: 109 numeric/encoded features derived from the 46 original + 22 engineered columns (raw text/ID/high-cardinality string columns excluded from the numeric model matrix but retained in the full processed dataset).